

Mohd Rizwan

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Applied AI / full-stack engineer with production experience shipping computer vision, LLM/RAG, and graph-database features across healthcare and research data platforms. Comfortable across the stack – Python, Django, Angular, Java/Spring Boot, REST APIs – with hands-on delivery of applied-AI systems (YOLO/R-CNN model training, LLM chatbots, RAG-style retrieval, Neo4j-based reasoning engines) end to end.

Work Experience

Software Engineer, MatheSis Labs – Noida, UP

Jan 2026 – Present

- Built full-stack features for a healthcare application using **Angular, HTML, CSS, JavaScript, and Python**, serving **1,000+** daily active users.
- Trained and deployed **computer vision models (YOLO, R-CNN)** for automated dental disease detection, reaching up to **95% accuracy**.
- Built and integrated an **LLM-based chatbot** with prompt design and response grounding (RAG-style retrieval) to improve user engagement.
- Designed a **Neo4j graph database** rule engine for differential diagnosis, containerized with **Docker** and exposed via **REST APIs**.
- Co-built a **Django, PostgreSQL, Redis** dataset-publishing platform (3-engineer team): full-stack upload/validation/publish workflow with a **Python ETL** de-identification pipeline and **Celery**-driven dataset versioning.
- Built interactive UI modules (including a gamified feature) and iterated on **UI/UX** based on clinical-user feedback.

Developer Intern, MatheSis Labs – Noida, UP

Jul 2025 – Dec 2025

- Built backend features using **Java, Spring MVC, and MySQL** across **5+** product modules.
- Developed and tested **REST APIs** and supported database design/queries using **SQL**.
- Followed **SDLC** practices under senior engineer review; delivered features within sprint timelines as part of a cross-functional team.

Analyst, HCLTech – Noida, UP

Nov 2022 – Mar 2024

- Supported large-scale enterprise applications, maintaining **96% SLA compliance** and operational stability.
- Diagnosed application logs, database issues, and service failures to improve **system reliability**.
- Partnered with development and support teams to cut recurring incidents by **25%**.

Projects

AI-based Healthcare Application, MatheSis Labs

- Refined **Angular** UI components and data flow between frontend and backend services to improve performance and usability.
- Built an end-to-end **CBCT/DICOM 3D visualization** pipeline for dental disease detection using **YOLO** on volumetric imaging data.

Research Dataset Publishing Platform, MatheSis Labs

- Built a full-stack **Django, PostgreSQL, HTML/CSS** platform for secure dataset and metadata-CSV upload, validation, and public release.
- Designed a **Python ETL** de-identification pipeline, verifying each validation checkpoint method-by-method; used **Celery and Redis** for async processing and dataset versioning.

Expense Tracker, Personal

- Built backend components in **Java, Spring Boot, and MySQL** with complete **CRUD** operations.
- Designed clean, maintainable **REST APIs** following **SOA** principles, validated with automated tests under concurrent access.

Technical Skills

- **Languages:** Java, Python, JavaScript, TypeScript, HTML, CSS
- **Applied AI:** LLM integration, prompt engineering, RAG-style retrieval, YOLO/R-CNN model training & deployment, computer vision
- **Frameworks:** Spring Boot, Spring MVC, Django, Angular, REST APIs, System Architecture, Microservices
- **Databases & Data Engineering:** MySQL, PostgreSQL, SQL, Neo4j (Graph DB), Redis, Celery (async task queues)
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, Maven, CI/CD
- **Core CS:** Data Structures & Algorithms, OOD, API Design, Concurrent Systems
- **Practices:** Agile/Scrum, Code Reviews, End-to-End Delivery, Debugging

Education

Maulana Azad National Urdu University, B.Tech in Computer Science

Jul 2018 – Jun 2022

- CGPA: 9.13 / 10.0

Achievements

- Ranked in the **top 3** throughout school and college for academic performance.
- **Co-founder, College Coding Club** – organized coding competitions and peer-mentorship sessions.
- **Gold Medal**, school-level Mathematics Competition; **1st Rank**, district-level Speech Competition.